PREPARED BY :	REVIEWED BY :	APPROVED BY :	
RIKI DEPANO (ENGINEERING SPV)	**PURNO BUDI KISWANTO (ENGINEERING MANAGER)**	**HAPPY MONDA PINTAULI (QA MANAGER)**	**BHIRAWA SEPTARIYANTO (PROJECT LEADER)**

1. TUJUAN

PURPOSE

Memberikan petunjuk tentang pengoperasian dan perawatan terhadap sistem pemantauan lingkungan dan *system* pemantauan gedung agar beroperasi dengan baik dan sesuai dengan standar yang ada di PT Etana Biotechnologies Indonesia.

Provides instruction about operating and maintaining of environment monitoring system, and building monitoring system to confirm the system running well base on standard in PT. Etana Biotechnologies Indonesia.

2. RUANG LINGKUP

SCOPE

SOP ini digunakan oleh departement *Engineering* khususnya teknisi *EMS & BMS* sebagai pedoman dalam melakukan pengoperasian dan perawatan terhadap sistem sistem pemantauan lingkungan dan *system* pemantauan gedung diarea Produksi, Warehouse & CUB.

This SOP is used by Engineering department especially EMS & BMS technician as guidance in order to operating and maintaining of environment monitoring system, cold storage monitoring and building monitoring system in the Production, Warehouse & CUB areas.

3. TANGGUNG JAWAB

RESPONSIBILITY

3.1. Teknisi *EMS & BMS* bertanggung jawab :

EMS & BMS technician are responsible :

3.1.1. Mengoperasikan sistem sesuai dengan kebutuhan.

Operate the system as needed.

3.1.2. Melakukan perawatan sesuai jadwal yang telah ditetapkan.
Perform maintenance according to a predefined schedule.

3.1.3. Melakukan pemantauan harian terhadap sistem.
Perform daily monitoring of system.

3.1.4. Melaporkan setiap ada penyimpangan kepada Supervisor *Engineering*.
Report any deviation to Engineering Supervisor.

3.2. Supervisor *Engineering* bertanggung jawab :
Engineering Supervisor is responsible :

3.2.1. Membuat jadwal perawatan secara berkala.
Create schedule of maintenance periodically.

3.2.2. Memastikan pelaksanaan perawatan sesuai dengan jadwal.
Ensure implementation of maintenance is according to the schedule.

3.2.3. Memastikan semua alat ukur yang terpasang dan digunakan telah sesuai dengan spesifikasi yang telah ditentukan (dikalibrasi).
Ensure that all measuring instruments installed and used are in accordance with the specifications (calibrated).

3.2.4. Memberikan pelatihan oprasional *EMS* kepada departemen lain yang terkait.
Provide operational EMS training to other departments.

3.2.5. Memastikan ketersediaan part kritikal dan jumlahnya cukup.
Ensure availability of critical parts and sufficient quantities.

3.2.6. Mengkoordinasikan dan melaporkan kepada Manajer *engineering* jika terdapat kerusakan pada sistem.
Coordinate and report to engineering manager if there is malfunction to system.

4.1. Manajer *engineering* bertanggung jawab :
Engineering manager responsible :

3.3.1. Menentukan *spare part*s pengganti apabila *spare part*s yang sesuai sudah tidak dijual lagi.
Determine spare parts if suitable spare parts are no longer sold.

3.3.2. Memutuskan dan mengkoordinir tindakan yang diperlukan, apabila perlu dilakukannya penggantian, modifikasi peralatan, perubahan sistem.
Decide and coordinate action required, if necessary replacement, equipment modification, change system.

4. DEFINISI

DEFINITION

4.1. *Environmental Monitoring System (EMS)* adalah suatu perangkat yang mampu melakukan monitoring kondisi lingkungan atau ruangan tertentu.

Environmental Monitoring System (EMS) is a device capable of monitoring the conditions of the environment or a critical room.

4.2. *Building Management System (BMS)* atau dikenal sebagai *Building Automation System (BAS)* adalah sistem kontrol berbasis komputer yang dipasang di gedung yang berfungsi untuk mengontrol dan memonitoring peralatan mekanis seperti *HVAC* dan sarana penunjang lainnya.

Building Management System (BMS) is a computer-based control system installed in a building that serves to control and monitor mechanical equipment such as HVAC and other support facilities.

5. PETUNJUK UMUM

GENERAL INSTRUCTION

5.1. Cara pengoperasian harus dengan benar atau sesuai urutan-urutannya dan perawatan harus dilakukan dengan teratur dan sesuai dengan jadwal yang telah dibuat.

The way of operation should be correct or in accordance with its sequence and maintenance should be done regularly and in accordance with the schedule that has been made.

5.2. Operator/teknisi yang telah terlatih atau sudah mendapatkan pelatihan dan terkualifikasi yang dapat mengoperasikan dan melakukan perawatan sistem *BMS & EMS*.

Trained or qualified training operator/technicians who can operate and conduct maintenance of EMS & BMS system.

5.3. Keselamatan kerja harus tetap di perhatikan selama pengoperasian atau perawatan.

Safety must be kept in mind during the operation or maintenance.

5.4. Kegiatan perawatan yang dilakukan secara berkala harus didokumentasikan setelah perawatan dilakukan.

Regular maintenance activities should be documented after maintenance done.

5.5. Apabila alat ukur tersebut didapati dalam keadaan rusak, harus diberi penandaan yang jelas.

If the measuring device is found to be in a defective state, it should be clearly marked.

5.6. Perlengkapan kerja sepatu *safety* dan helm harus digunakan.

Work equipment safety shoes and helmet must used

5.7. Setelah melakukan perawatan, semua kegiatan harus didokumentasikan lengkap pada Formulir Perawatan Berkala *EMS* dan *BMS* No. SOP/EBI/ENG-055-F01.

After conduct maintenance, all activities must be fully documented in the form Periodic Maintenance of EMS and BMS No. SOP/EBI/EN-055-F01.

5.8. Semua kegiatan perbaikan ataupun perawatan yang dilakukan harus dicatat pada Formulir Kartu Riwayat Mesin *No. SOP/EBI/EN-055-F01.*

All repairs or maintenance activities carried out must be recorded on the machine history card Form No. SOP/EBI/EN-055-F01.

5.9. Setiap suku cadang yang datang dari supplier harus dicek keaslian, dimensi serta kesesuaian spesifikasinya.

Every spare part that comes from the supplier must be checked for authenticity, dimension and suitability of the specification.

5.10. Penggantian suku cadang harus sesuai dengan spesifikasi dan fungsinya. Untuk komponen kritis atau suku cadang yang berhubungan langsung dengan produk apabila diganti dengan beda *merk* dan *type* yang berbeda tetapi satu jenis maka dibuatkan *change control*.

Spare parts replacement must be in accordance with the specifications and functions. For critical components or spare parts that are directly related to the product if replaced with different brands and different types but one type, then change control is made.

5.11. Untuk perubahan major yang mendasar dan merubah sistem harus memakai formulir *change control* dan disetujui oleh manajer terkait.

For major underlying changes and changing the EMS & BMS system must use a change control form and be approved by the relevant manager.

5.12. Untuk akses *user personel* telah ditentukan menjadi 3 bagian, yaitu :

For user personnel access, it has been determined to be 3 parts, namely:

Operator, Digunakan oleh operator/teknisi untuk melakukan pengoperasian pada *BMS & EMS.*

5.12.1 Pengoperasian, Digunakan oleh operator/teknisi untuk melakukan operasional pada *BMS & EMS.*

Operator, Used by operators/technicians to perform operations on the BMS and EMS.

5.12.2 Supervisor, digunakan oleh Supervisor terkait untuk merubah parameter pada *BMS & EMS*.

Supervisor, Used by related Supervisors to change parameters in BMS and EMS.

5.12.3 Admin, digunakan oleh Manajer untuk merubah parameter, unduh riwayat *Alarm* dan menambahkan *user* untuk operator dan *maintenance BMS & EMS*.

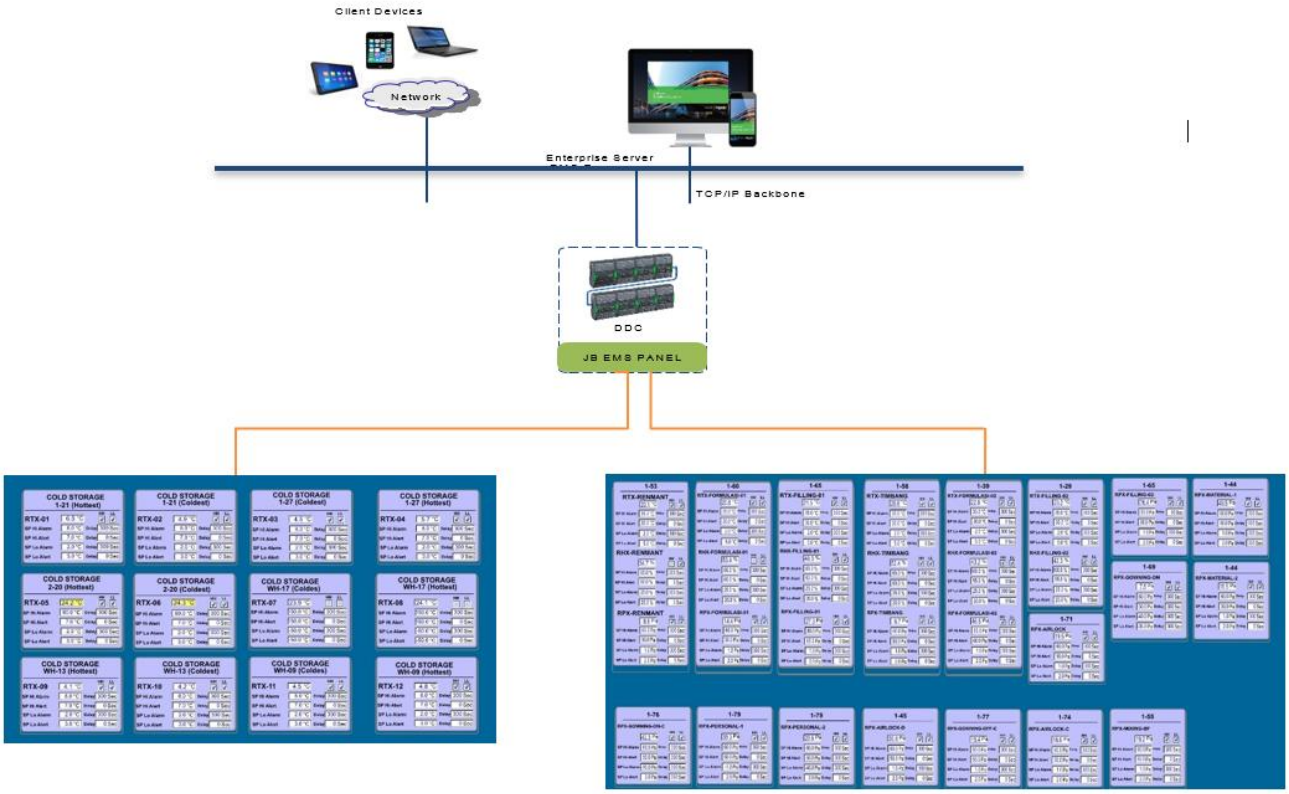
Admin, Used by Manager to change parameters, download Alarm history and add user for BMS and EMS. operator and maintenance.

5.12.4 *Password* rutin dirubah tiap 3 bulan.

Password are routinely changed every 3 months.

5.12.5 Topologi *EMS & BMS* pada ETANA terdiri dari beberapa komponen, diantaranya sebuah *Enterprise Server, Juntion Box DDC Panel, temperature sensor, pressure sensor, dan humidity sensor.*

The EMS & BMS topology on ETANA consists of several components, including an Enterprise Server, Juntion Box DDC Panel, themperature sensor, pressure sensor, and humidity sensor.



Gambar 1 Topologi *EMS & BMS*

Figure 1 Topology EMS & BMS

5.12.6 *EMS* digunakan untuk monitoring *temperature*, RH, dan DP area kritikal di ruang produksi dan *cold storage. BMS* digunakan untuk control sistem *HVAC* area produksi. *EMS is used to monitor critical temperature, RH, and DP areas in manufacturing and cold storage facilities. BMS is used to control the HVAC system in the production area.*

6. ALAT DAN BAHAN

TOOL AND MATERIAL

6.1. **ALAT**

TOOLS

6.1.1. Kain Lap

Duster

6.1.2. Tang ampere

Ampere meter

6.1.3. Tang

Pliers

6.1.4. Obeng

Screw

6.2. **BAHAN**

MATERIAL

Tidak ada

None

7. PROSEDUR

PROCEDURE

7.1. Pengoperasian *EMS & BMS*.

Operation of EMS & BMS.

Pengoprasian *EMS & BMS* dapat di lakukan dengan 2 cara yaitu *Workstation* atau melalui *Webstation*

The two methods for EMS and BMS communication are through a Workstation and a web station, respectively.

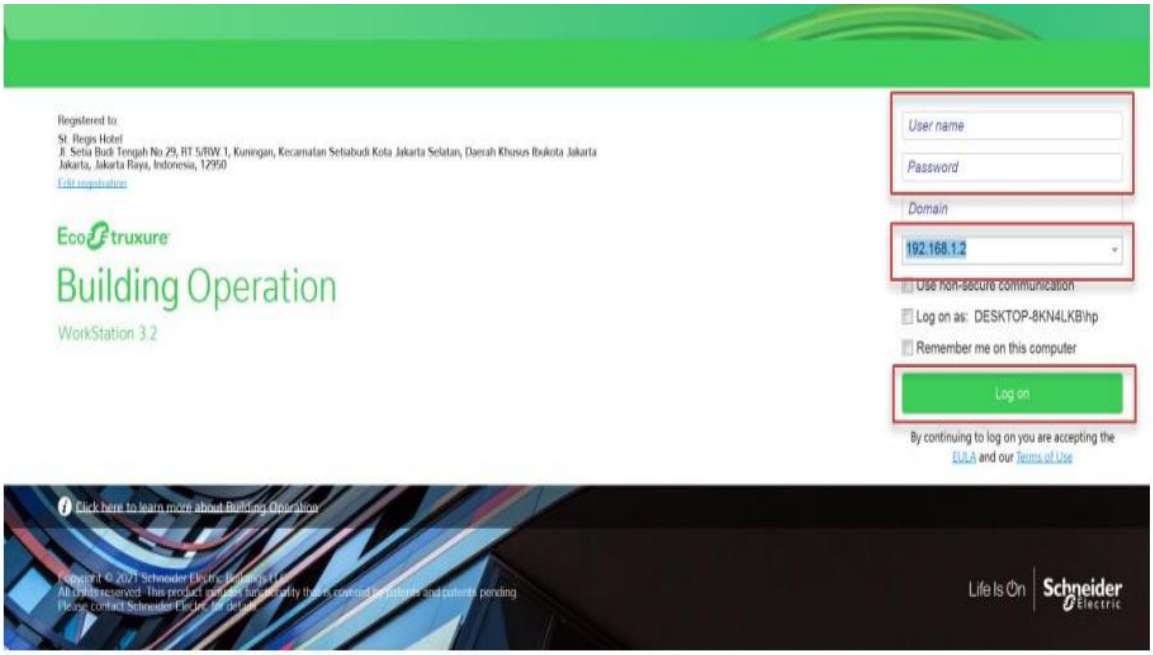
7.1.1. *WORKSTATION*

WORKSTATION

Workstation merupakan *software* yang dikembangkan oleh *Schneider electric* untuk pengoperasian dan pemrograman *EMS & BMS Schneider*. Langkah untuk *log in* menuju *Workstation* adalah dengan membuka aplikasi *Workstation building operation 3.2*, dan kemudian masukan *username* dan *password* yang telah disiapkan oleh *BMS Engineer* seperti Gambar 2 dibawah. Pastikan targetnya ke IP *Enterprise Server*, kemudian lanjutkan dengan klik tombol *log In.*

Workstation is a piece of software that Schneider Electric developed for use with their EMS and BMS programs. The procedure for logging into a Workstation is to launch the Workstation building operation 3.2 application, then enter the username and

password that the BMS Engineer predetermined as shown in the table below. Paste your target at the IP Enterprise Server, and then move forward by clicking the log-on button.



Gambar 2 *Workstation Log in*

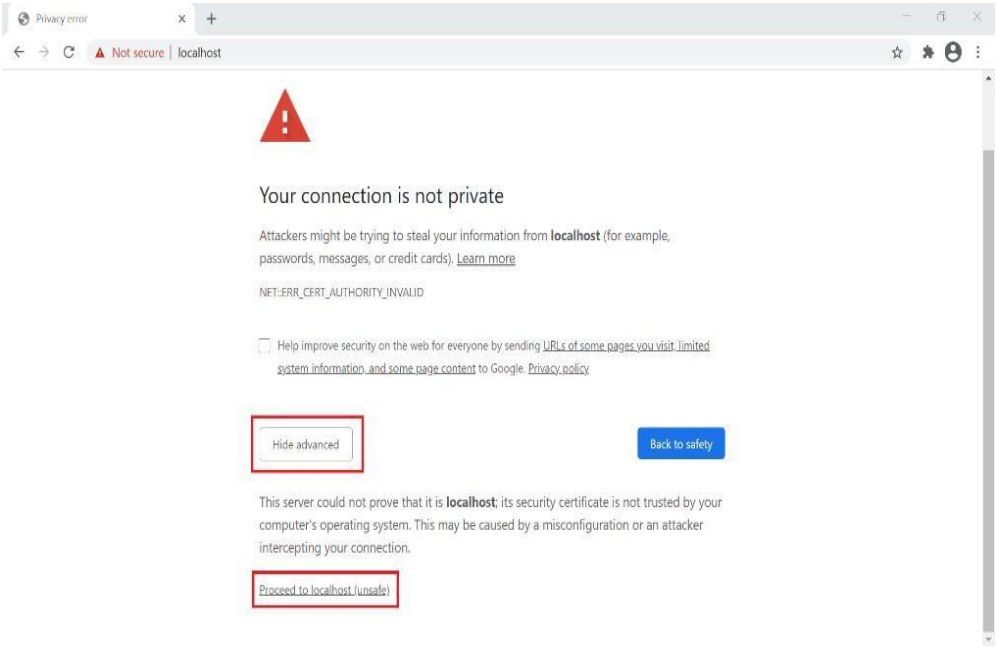
Figure 2 Workstation Log in

7.1.2. *WEBSTATION*

WEBSTATION

Webstation merupakan *station EMS & BMS* yang dapat diakses melalui *web browser*. Langkah untuk mengakses *Webstation* adalah dengan membuka *web browser*, kemudian ketikkan *IP Address Enterprise Server* pada kolom pencarian *browser*, maka akan terbuka halaman seperti pada Gambar. Klik tombol *proceed to localhost (unsafe).*

Webstation is an EMS and BMS station that can be accessed through a web browser. To access the web station, start a web browser and then select the IP Address Enterprise Server from the browser menu bar. Will open the page as window in the figure below. Click proceed to local host (unsafe) botton.



Gambar 3 *Acces Webstation*

Figure 3 Acces Webstation

7.1.3. *LOG IN*

LOG IN

Masukkan *username* dan *password* pada halaman yang tersedia sesuai dengan parameter yang diberikan oleh *EMS & BMS Engineer*.

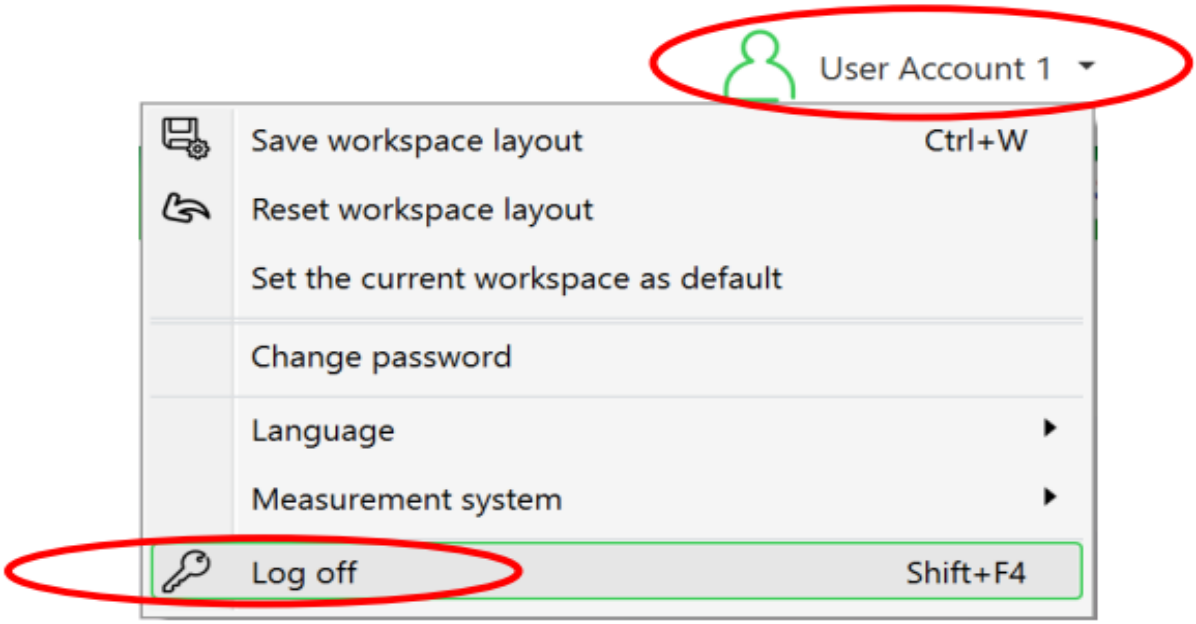
Enter your login and password in the field provided on the available page in accordance with the specification provided by the EMS & BMS Engineer.

7.1.4. *LOG OFF*

LOG OFF

Anda dapat keluar dari *Workstation/Webstatation* sehingga personel yang tidak berwenang tidak dapat merusak pengaturan Anda. Keluar atau *Log-off* tidak berarti Anda menutup aplikasi *Workstation*. Untuk keluar dari *Workstation* Dalam *Workstation*, dalam menu klik *log off user account*, klik *Log Off.*

You can leave the Workstation/Webstatation so that uncooperative staff members cannot question your qualifications. Keluar atau Log-off does not mean that you are starting the Workstation application. To depart from the Workstation Select Log Off from the Workstation's menu under the account logo.



Gambar 4 *Log Off*

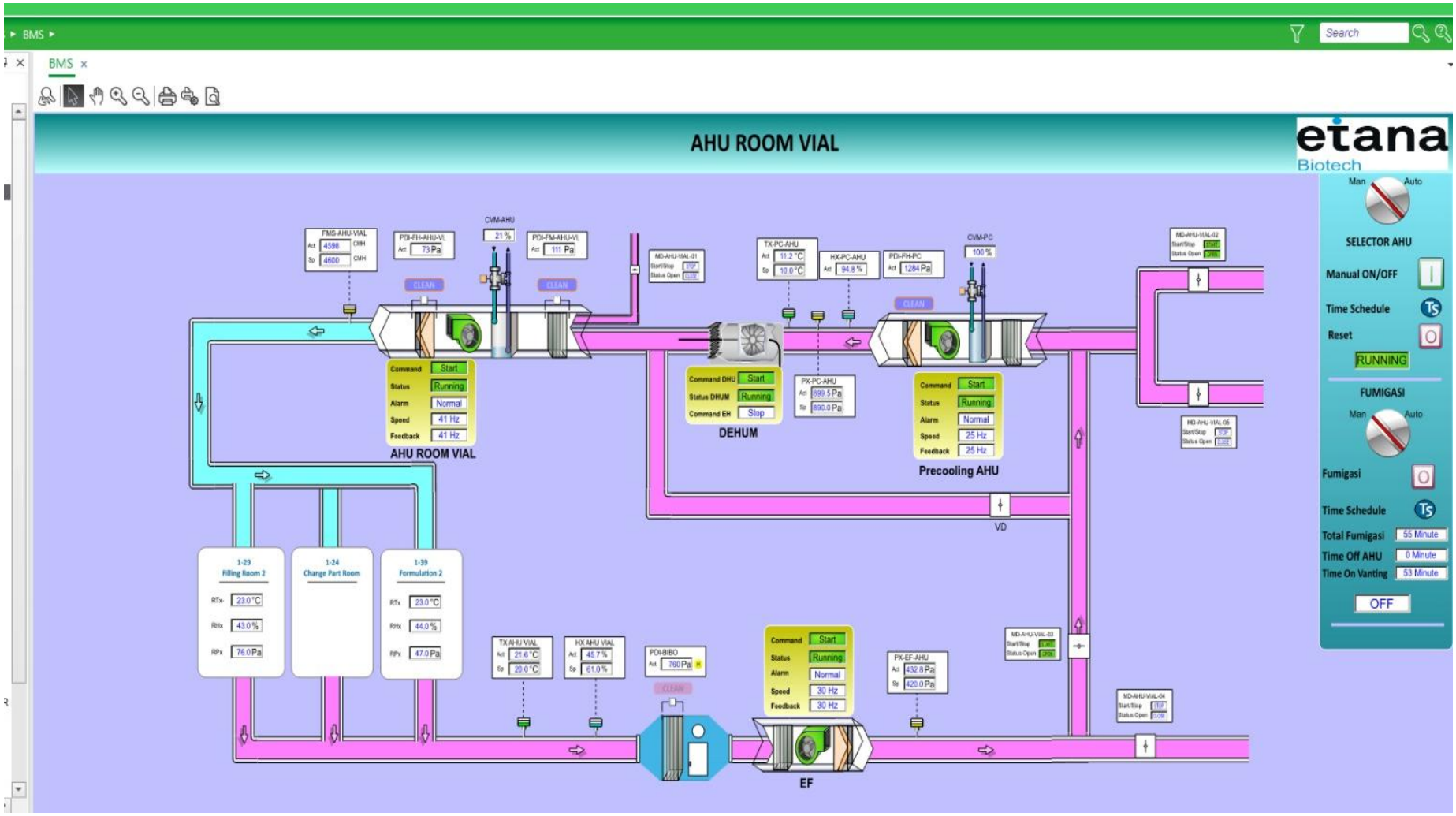
Figure 4 Log off

7.1.5. *Monitoring* Harian *EMS & BMS*

Daily Monitoring EMS & BMS

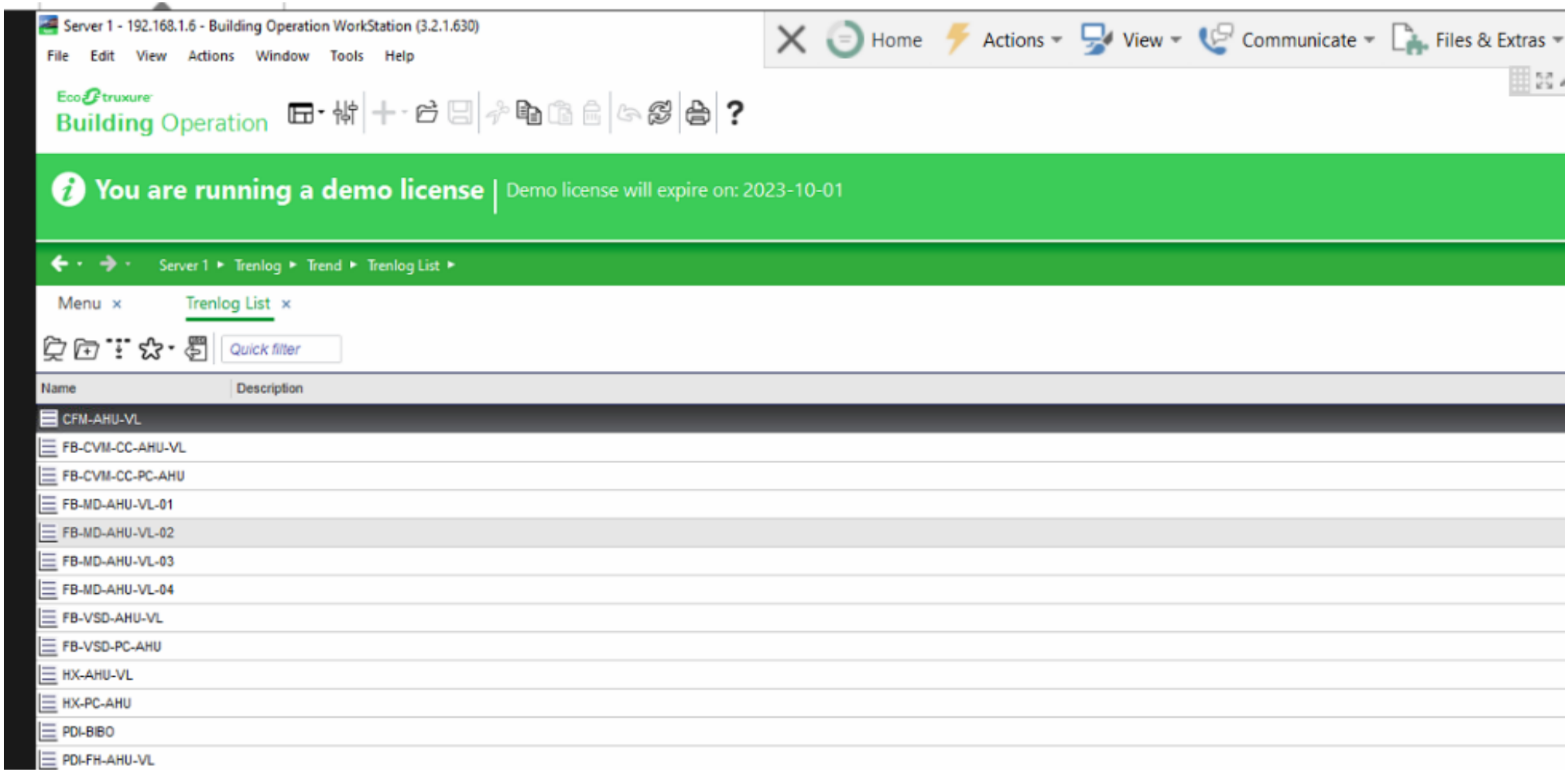
Trenlog ini untuk me*record* nilai yang di monitoring baik itu point *Digital Input, Digital Output, Analog Input and Analog Output*. Disini akan me*record* nilai yang dibaca pada monitoring untuk lamanya waktu *record* bisa kita sesuaikan sesuai kebutuhan kita. Trenlog *list* datanya juga dapat di *Export* langsung seperti gambar no. 4 Trenlog ini fungsinya untuk melihat apakah *system* me *record* dengan baik data *temperature*, *humidity* dan *differential pressure*. Dengan mengklik trenlog

This is a Trenlog to record the values that are being monitored, whether it's the point Digital Input, Digital Outout, Analog Input and Analog Output. Here it will record the value read on the monitoring for the length of time the record can be customized according to our needs. The Trenlog data list can also be Exported directly as figur 4 The Trenlog function is to see if the system is recording temperature, humidity and differential pressure data properly. By cliking trenlog button



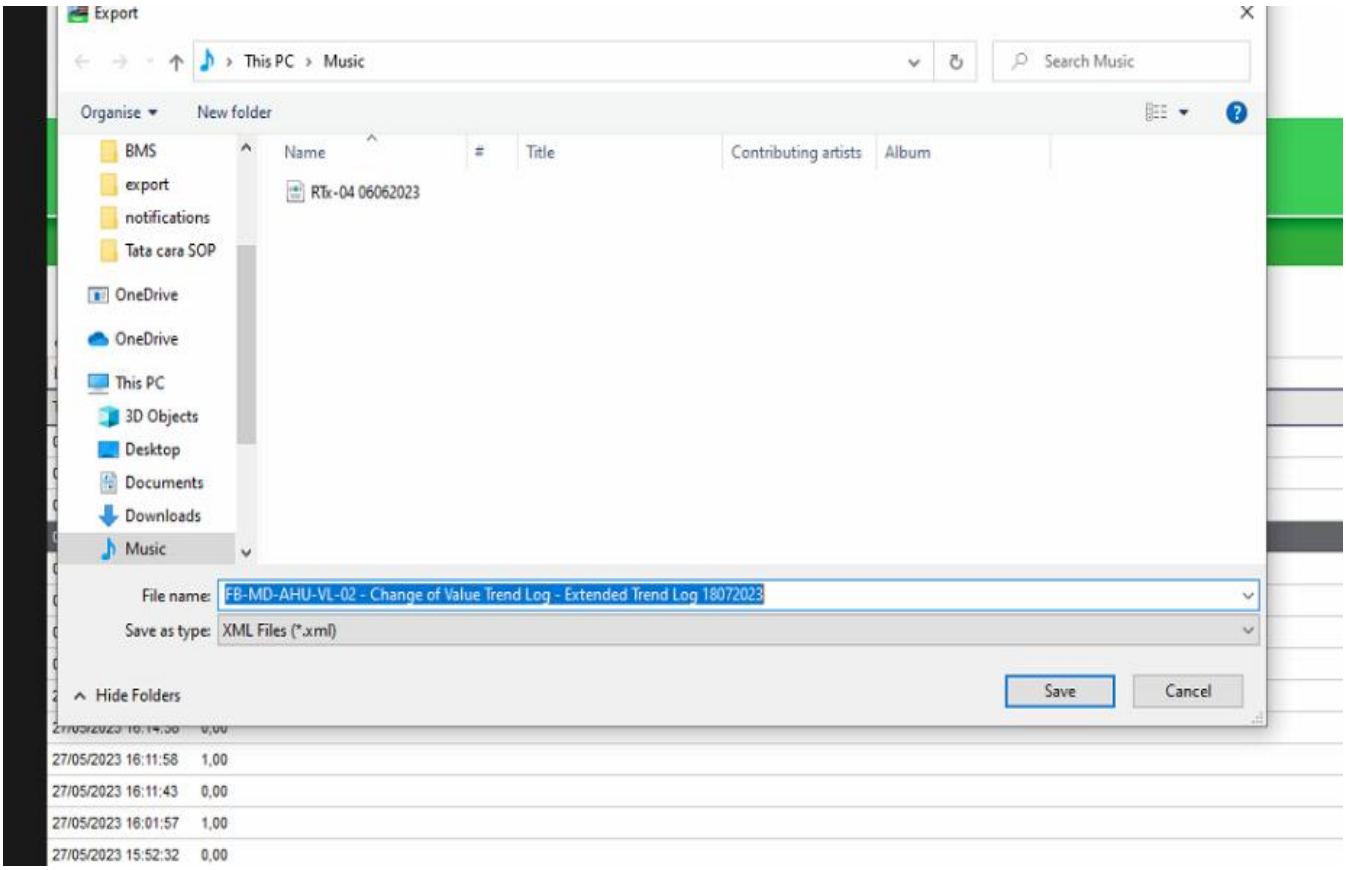
Gambar 5 Monitoring display AHU ROOM VIAL

Figure 5 Monitoring display AHU ROOM VIAL



Gambar 6 Monitoring *trenlog list*

Figure 6 Monitoring trenlog list



Gambar 7 *Export trendlog list*

Figure 7 Export trendlog list

7.1.6. *List Alarm Audit Trial EMS & BMS*

List Alarm Audit Trial EMS & BMS

Audit Trail Ini berfungsi untuk melihat aktifitas *user* apa kegiatan *user* dan perubahan yang dilakukan akan sesuai kebutuhan lapangan, kita juga bisa melihat kapan *user* aktif untuk melakukan segala perubahan yang dilakukan oleh *user* akan terekam pada *Audit Trail***.** Untuk membuka *Audit Trail* gunakan akun *Manager* :

The purpose of the Audit Trail is to monitor user activity, including what projects user are working on and what changes user are making to meet user needs. User can also observe when users become active to make changes, as all changes made by users will be recorded in the Audit Trail. To open an audit trial use manager account:

7.1.6.1. *Login User admin*

Login User admin

7.1.6.2 Klik *Server*

Klik *Server*

7.1.6.3 Klik *control Panel*

Klik *control Panel*

7.1.6.4. Pilih *User Management*

Choose User Management

7.1.6.5. Pilih *Acount*

Choose Account

7.1.6.6. *View All account* dan pilih salah satu *account* yang mau dilihat *Audit Trail*.

View All Account and select the account that requires an Audit Trail.

7.1.6.7 Klik *Audit Trail* Event

Klik *Audit Trail Event*

7.1.7. Penarikan Data *Record EMS & BMS*

Recording EMS and BMS Data

7.1.7.1 Ketik *windows + run*

Click windows + run

7.1.7.2 Ketik administrator

Click administrator

7.1.7.3 Klik *folder notification*

Click folder notification

7.1.7.4 Pilih data hasil *record* yang akan diperlukan

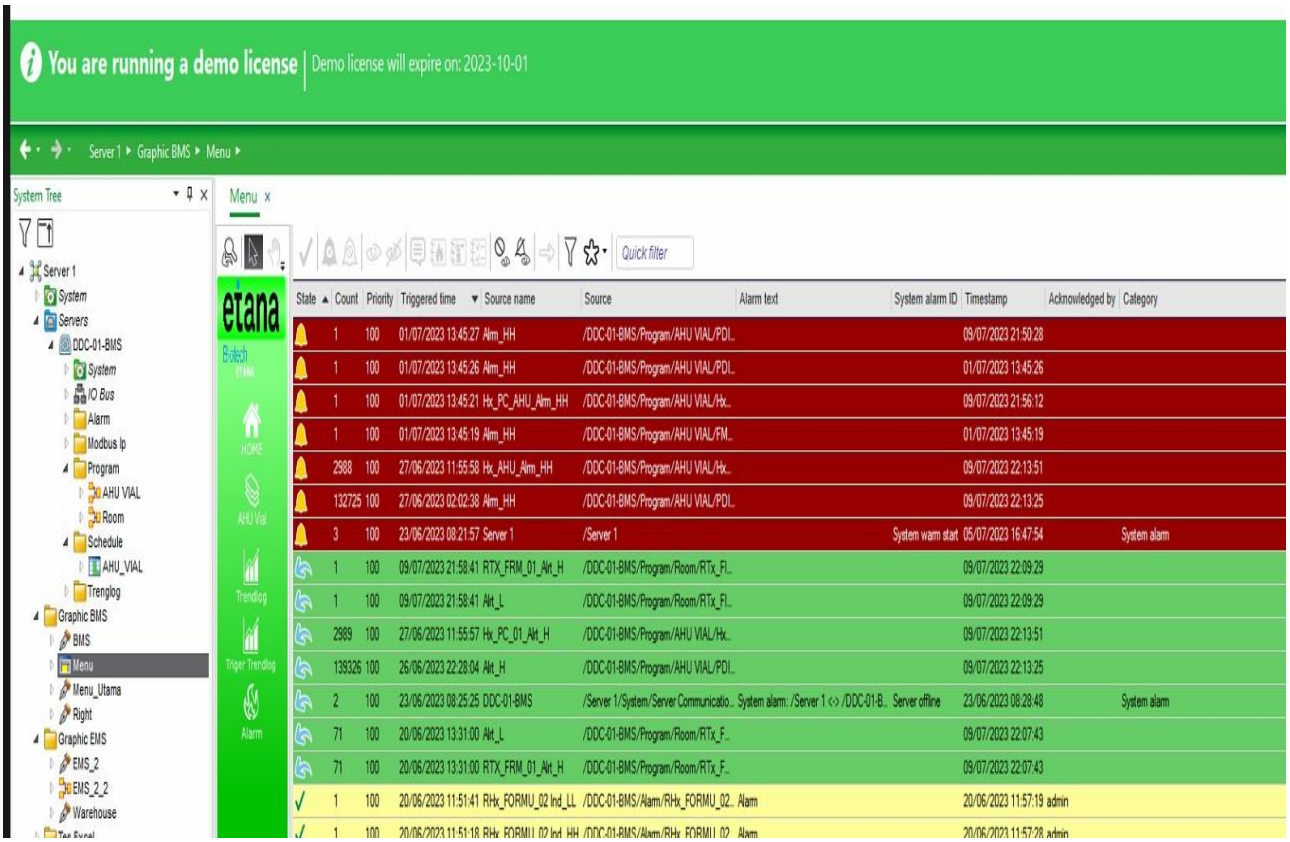
Choose data record which needed

7.1.8. *Alarm BMS & EMS*

Alarm BMS & EMS

Alarm ini menampilkan *trip* atau batas yang kita anggap tidak wajar, seperti suhu terlalu tinggi dan ada juga kondisi lapangan seperti motor sedang ada perbaikan. Apabila *AHU* tidak hidup berdasarkan urutan maka akan terjadi *trip* juga (*Mismatch*) itu bisa juga dikarenakan ada panel yang sedang di manual sehinga mengakibatkan *Mismacth*, ini untuk mencegah kerusakan peralatan juga berikut dibawah contoh tampilan *Alarm BMS*.

This Alarm showing Trips or limits that we think are unreasonable, such as too high temperatures and there are also field conditions like the motorcycle is being repaired. If the AHU does not live according to the sequence then there will be a Trip too (Mismatch) it can also be because there are panels in the manual as long as resulting Mismacth, this to prevent equipment damage also follows below the example of the display of the BMS Alarm.



Gambar 8 *Alarm EMS & BMS*

Figure 8 Alarm EMS & BMS

7.1.9. Pengoperasian *BMS AHU*

Operational BMS AHU

Pengoperasian *AHU* dapat dilakukan dalam mode manual dan otomatis dengan cara memilih selektor pada mode manual atau otomatis pada *display*.

AHU operations can be done in manual and automatic mode by selecting selectors in manual mode or automatic on display.

7.2. Perawatan berkala 1 bulanan (L1), 3 bulanan (L2), dan L4 tahunan

Periodic maintenance of 1 month (L1), 3 months (L2), and yearly (L4)

7.2.1. Perawatan 1 bulanan (L1).

1 month maintenance (L1).

7.2.1.1. Memastikan seluruh panel DDC sudah berjalan dengan baik yaitu lampu indikator status pada setiap modul DDC berwarna hijau dan bukan berwarna merah, serta lampu indikator *ethernet* hijau.

Ensure that all of the DDC panels are operating properly by ensuring that the status indicators on each DDC module are green and not red, as well as the high-visibility ethernet indicator bulb.

7.2.1.2. Pastikan pulsa dan masa aktif pada sim card tersedia

Create sure the pulse and active time on the sim card is available

7.2.1.3. Kencangkan koneksi kabel pada komponen dan terminal

Tighten cable connections to components and terminals

7.2.1.4. Periksa kondisi MCB, pastikan MCB dalam keadaan baik.

Check the condition of the MCB, create sure it's in good shape

7.2.1.5. Periksa ruang penyimpanan pada hard disk komputer *EMS* dan *BMS*.

Check the storage space on the hard disk of EMS and BMS computers.

7.2.2. Perawatan 3 bulanan (L2).

3 month maintenance (L2).

7.2.2.1. Periksa kondisi badan panel dari kelembaban dan debu yang mungkin masih dapat masuk ke dalam panel dan bersihkan.

Check the condition of the panel body from moisture and dust that may still be able to enter the panel and clean it.

7.2.2.2. Periksa koneksi dan komponen – komponen kelistrikan pada panel DDC. Pastikan kondisi komponen tidak rusak dan ganti jika diperlukan.

Check the connections and electrical components on the DDC panel. Create sure the condition of the components is intact and replace if necessary.

7.2.2.3. Periksa tegangan output power supply, pastikan tengangannya lebih dari 24V dan ganti komponen apabila output power supply < 20V

Check the output voltage of the power supply, create sure the voltage is more than 24V and replace the components when the power output is < 20V

7.2.2.4. Cek koneksi kabel *LAN* pada *HUB*, pastikan koneksi kencang lampu indikator hijau menyala

Check the LAN cable connection on the HUB, create sure the green indicator light is on.

7.2.3. Perawatan 1 tahunan (L4).

7.2.3.1. Memastikan software yang diperlukan untuk menjalankan *EMS & BMS* sudah terinstal dan statusnya *running* pada PC *Central Management*. *Software* yang diperlukan terdiri dari: *License Administrator Building*

Operation 3.2, Software Administrator Building Operation 3.2, dan *Workstation Building Operation 3.2.*

Ensure that the software required to run EMS & BMS is already installed and running on PC Central Management. Required software consists of: License Administrator Building Operation 3.2, Software Administrator Construction Operation 3.2 and Workstation Building Operations 3.2

7.2.3.2. Kencangkan koneksi kabel pada komponen dan terminal.

Tighten cable connections to components and terminals

7.2.3.3. Periksa kondisi MCB, pastikan MCB dalam keadaan baik.

Check the condition of the MCB, create sure it's in good shape.

7.2.3.4. Periksa ruang penyimpanan pada hard disk komputer *EMS* dan *BMS*, lakukan backup data tahunan.

Check the storage space on the hard disk of EMS and BMS computers, create annual data backups.

8. REFERENSI

REFERENCE

8.1. *Manual Operation Environtment Management System.*

8.2. *Manual Operation Building Management system.*

9. LAMPIRAN

ATTACHMENT

9.1. *Logbook* Pemantauan Harian terhadap Sistem *EMS & BMS* No. SOP/EBI/EN-055-F01A.
Daily Monitoring of EMS & BMS Form No. SOP/EBI/EN-055-F01A.

9.2. Formulir Perawatan Berkala terhadap *BMS & EMS* No. SOP/EBI/EN-055-F02A.
Periodic Maintenance of BMS and EMS No. SOP/EBI/EN-055-F02A.

10. RIWAYAT

HISTORY

Tanggal ***Date***	**No. *Change control*** ***Change control No.***	**No. Revisi** ***No. Revision***	**Alasan Perubahan** ***Reason of Change***
	CC-EBI-EN-005-23 & CC-EBI-QA-037-23	00	Dokumen baru *New Document*

11. DISTRIBUSI

DISTRIBUTION

11.1. Departemen Pemastian Mutu

Quality Assurance Department

11.2. Departemen *Engineering*

Engineering Department

11.3. Departemen Produksi

Production Department

11.4. Departemen Warehouse

Warehouse Department

etana Biotech **PT ETANA BIOTECHNOLOGIES INDONESIA**	**LOGBOOK / *LOGBOOK*** **PEMANTAUAN HARIAN TERHADAP SISTEM *EMS* dan *BMS*** ***DAILY MONITORING OF BMS & EMS***	Document No. : SOP/EBI/EN-055-F01A Effective Date :

Date :

Checklist item	Standard Value	Time (Shift 1, Shift 2, Shift 3)			Remarks
		10:00	16:00	03:00	
Check UPS	Running				
Check *record EMS*	*Record*				
Check *Alarm* list (Upper limit and lower limit)	Normal				
Check parameter terkontrol	Normal				
Cleaning monitor dan CPU	Clean				
Note :	Operator Shift 1 ( ) Shift 2 ( ) Shift 3 ( )				Checked By : ( )

etana Biotech **PT ETANA BIOTECHNOLOGIES INDONESIA**	**FORMULIR / *FORM*** **PERAWATAN BERKALA TERHADAP *BMS & EMS*** ***PERIODIC MAINTENANCE OF BMS & EMS***	Document No. : SOP/EBI/EN-055-F02A Effective Date :

Level	:
Month	:
Date	:
Day	:

Level	Component	Activity							Condition		Remark
		C	I	L	R	S	T	W	B	G	
L1											
	Indicator lamp										
	Pulsa										
	All cable terminal connection										
	Inspection all circuit breaker										
	Data storage										
L2											
	Electrical panel										
	Power supply 24V										
	LAN connection										
L4											
	Backup data *record*										
	License Administrator Building Operation										
	Software Administrator Building Operation										
	Workstation Building Operation										
	Enterprise Server										

Note :

No	Date	Start	Finish	Duration		Total personnel	Area	Remark
				Hour	Minute			

Remark : C : Cleaning (pembersihan) I : Inspection (pemeriksaan) L : Lubrication (pelumasan) R : Replacement (penggantian) S : Setting (penyetelan) T : Tightened (dikencangkan) W : Welding (pengelasan) G : Good (baik) B : Bad (jelek)	Level : L1 : Every 1 month L2 : Every 3 month L4 : Every 1 year	Done by : ( )	Checked by : ( )